

From ξ to φ : Recursive Scale Hierarchy

Spectral Gap Structure of FFGFT T^4 and HLV G-Lattice

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Abstract

We show that the golden ratio φ in FFGFT is not an independent parameter, but the fixed point of a recursive damping sequence that follows solely from the fundamental geometric parameter $\xi = 4/30000$ and the fractal dimension $D_f = 3 - \xi$. After $k^* = \log(\varphi)/\xi \approx 3609$ recursion levels one obtains $r(k^*) \approx 1/\varphi$ with an absolute error of $4 \cdot 10^{-5}$ (relative $6.5 \cdot 10^{-5}$) at integer $k = 3609$; the exact real $k_{\text{exact}}^* = 3608.51$ (with the $r(0)$ correction) hits $1/\varphi$ identically. The recursion levels are not renormalisation-group steps with factor 2, but geometric scale levels with the FFGFT-internal factor $q = 2/3$; with this factor $(2/3)^{3609}$ is a convergent, theory-internal scaling — not an astronomical limit. This result connects the microscopic fractal structure of the T^4 -torus with the macroscopic φ -geometry appearing in the HLV lattice (6D→3D cut-and-project) without free parameters. A numerical validation study compares the spectral gap structure of FFGFT T^4 , HLV G-Lattice, and null models across three layers.

1 Motivation and Context

Doc. 258 showed that the Koide scalar $Q_{\text{FFGFT}} \approx 0.6677$ follows from the geometric invariants r_e, r_μ, r_τ and the lepton exponents p_e, p_μ, p_τ . This value is an *output* of the theory, not an input parameter.

In parallel, the HLV lattice construction of Krüger [1] uses the golden ratio $\varphi = (1 + \sqrt{5})/2$ as the projection ratio in the 6D→3D cut-and-project construction. Krüger's Run E identifies the low-sector gap hierarchy as the only remaining candidate separation channel between HLV and null models.

This document answers three questions:

1. Which scalar is physically correct for the macroscopic geometric structure?
2. How does it connect to ξ ?
3. What does $k^* \approx 3609$ mean physically — and why is it not an astronomical scale?

Remark .1.1 (All three FFGFT scalars are derivable). The question is not which of the three scalars $\xi, 1/\varphi, Q_{\text{FFGFT}}$ can be derived from FFGFT — that holds for all three. The question is which one *operates* at which physical level and is therefore responsible for which observables.

2 Fundamental Parameters and Scalar Separation

FFGFT works with the following parameter-free fundamental quantities:

$$\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4} \quad (\text{fractal deviation, microscopic}) \quad (1)$$

$$D_f = 3 - \xi \approx 2.99987 \quad (\text{Hausdorff dimension of } T^4) \quad (2)$$

$$q = \frac{2}{3} \quad (\text{geometric recursion factor}) \quad (3)$$

Definition .2.1 (Three FFGFT scalars near 0.6).

Quantity	Value	Role
$\xi = 4/30000$	1.333×10^{-4}	fundamental input; fractal dimension; active everywhere
$1/\varphi$	0.6180	macroscopic shell ratio; fixed point of ξ -recursion (output)
Q_{FFGFT}	0.6677	Koide scalar leptons (output, Doc. 258); particle physics scale only

All three are derivable from FFGFT. They are *not* interchangeable.

.3 The ξ -Damping Recursion**Definition and Closed Form**

Definition .3.1 (ξ -damping recursion). Let $r(0) = D_f/3 = (3 - \xi)/3$ be the shell ratio at the lowest scale level. The recursion reads:

$$r(k+1) = r(k) \cdot (1 - \xi), \quad k = 0, 1, 2, \dots \quad (4)$$

Closed form:

$$r(k) = \frac{D_f}{3} \cdot (1 - \xi)^k \approx \frac{D_f}{3} \cdot e^{-\xi k}. \quad (5)$$

The physical content: $r(0) = D_f/3$ is the ratio of the fractal dimension to the topological dimension 3. Each scale level damps this ratio by the factor $(1 - \xi)$.

Main Result: $1/\varphi$ as Fixed Point

Theorem .3.2 ($1/\varphi$ as fixed point of the ξ -recursion). *There exists a unique recursion depth*

$$k^* = \frac{\log \varphi}{\xi} = \frac{\log(\frac{1+\sqrt{5}}{2})}{4/30000} \approx 3609 \quad (6)$$

such that $r(k^) \approx 1/\varphi$ up to an absolute error of $4 \cdot 10^{-5}$ at integer $k = 3609$. The exact real $k_{\text{exact}}^* = (\log(1/\varphi) - \log(D_f/3)) / \log(1 - \xi) = 3608.51$ satisfies $r(k_{\text{exact}}^*) = 1/\varphi$ identically.*

Proof. From (5): $r(k^*) = 1/\varphi$ iff $(D_f/3) \cdot (1 - \xi)^{k^*} = 1/\varphi$. Since $D_f/3 \approx 1$ for small ξ : $(1 - \xi)^{k^*} \approx e^{-\xi k^*} \approx 1/\varphi$, hence $k^* \approx \log(\varphi)/\xi$. Numerical verification:

$$\begin{aligned} r(0) &= 0.9999\bar{5} \\ r(k=3609) &= 0.6179940013\dots \\ 1/\varphi &= 0.6180339887\dots \\ |r(3609) - 1/\varphi| &= 4.0 \times 10^{-5} \quad (\text{relative } 6.5 \times 10^{-5}). \quad \square \end{aligned}$$

□

Remark .3.3 (Parameter-free). $k^* = \log(\varphi)/\xi$ contains only $\xi = 4/30000$ (definition quantity) and φ (mathematical constant). No free parameter, no fit.

.4 Scale Interpretation: $q = 2/3$, Not Factor 2

Remark .4.1 (Critical point). A naive reading might interpret $k^* \approx 3609$ as astronomically many renormalisation-group steps: with factor 2 per step that would be a scale spread of $2^{3609} \approx 10^{1086}$, far beyond any experimental reach.

This objection misses the FFGFT structure.

The four torus lengths follow $L_\mu = L_0 \cdot q^\mu$ with $q = 2/3$. The cumulative scaling after k levels is $S(k) = (2/3)^k \rightarrow 0$.

Proposition .4.2 (Convergence with FFGFT scaling). *Comparison:*

Interpretation	Scale factor at $k^* = 3609$	Character
RG, factor 2 (external convention)	$2^{3609} \approx 10^{1086}$	divergent
FFGFT, $q = 2/3$ (internal)	$(2/3)^{3609} \approx 10^{-635}$	convergent

$k^* \approx 3609$ is a natural number within the theory's scale spectrum, not an astronomical limit.

.5 Spectral Validation Study (Three Layers)

Design

Compared: FFGFT T^4 (analytic Laplacian, ξ -damping, $q = 2/3$), FFGFT T^4 finite graph (kNN Laplacian on T^4 point grid, same method as HLV), HLV G-Lattice (NNLS-weighted kNN graph), and four null models (periodic torus, cubic, jittered-cubic, random-geometric). Script v3: $N = 2000$, $k = 16$, three seeds for FFGFT graph and HLV.

Layer 1 Gap-hierarchy metrics: max. gap, gap-CV, low-sector $\langle r \rangle$, IDS — analogous to Krüger Run E.

Layer 2 Shell structure: $1/\varphi$ vs. Q_{FFGFT} on the HLV point cloud at $N = 2000$.

Layer 3 Numerical verification of k^* and $r(k^*)$.

Remark .5.1 (Heuristics in the script). The shell-structure function uses a 5% threshold for shell detection. This is a heuristic; for a publication the radial distribution function $g(r)$ would be more robust. The ensemble size (3 seeds) is illustrative, not statistically secured.

Results Layer 1: Gap Hierarchy ($N = 2000$)

Three findings:

1. **FFGFT T^4 finite graph lies outside the null band** (gap_CV = 1.68 ± 0.09 , band upper bound 1.05). This is a stronger result than in v2.
2. **HLV lies inside the null band:** gap_CV = $0.755 \approx$ jittered-cubic = 0.755 (difference < 0.001). HLV is spectrally indistinguishable from the jittered-cubic control. This is *consistent* with Krüger's Run E.
3. **FFGFT analytic and HLV** both lie inside the band, but for different reasons: FFGFT analytic has a different spectral structure than the finite graph; HLV shows the same gap_CV as a random control.

Remark .5.2 (Consequence for the comparison). The finite FFGFT T^4 graph and HLV do not lie in the same gap_CV regime. This constrains the hypothesis of a common spectral signature formulated in Doc. 271: structural similarity is visible at the analytic level, but not robust at the graph level at $N = 2000$.

Table 1: Gap-hierarchy metrics, low sector ($n = 100$ modes, $N = 2000$). FFGFT graph: mean over 3 seeds with jitter $\sigma = 0.05$. Null band: $[0.31, 1.05]$ from the four null models.

Model	max_gap	gap_CV	low $\langle r \rangle$	Band status
FFGFT T^4 analytic	4.871	0.831	0.370	in band
FFGFT T^4 graph (mean $\pm$ std)	0.047	1.680 ± 0.092	0.452	outside
HLV G-Lattice	0.185	0.755	0.461	in band
Periodic torus	2.000	0.306	0.833	(boundary)
Cubic	0.250	1.051	0.343	(boundary)
Jittered-cubic	0.118	0.755	0.456	–
Random-geometric	0.105	0.616	0.567	–

Results Layer 2: Shell Structure ($N = 2000$)

At $N = 2000$ the discrimination between $1/\varphi$ and Q_{FFGFT} is measurable for the first time:

$$|\text{err}(1/\varphi) - \text{err}(Q_{\text{FFGFT}})|_{\text{HLV}} = |0.949 - 0.804| = 0.145 > 0.05. \quad (7)$$

$Q_{\text{FFGFT}} \approx 0.6677$ fits the HLV shell structure better than $1/\varphi \approx 0.6180$ at $N = 2000$. This is a *preliminary* finding — both values show absolute errors > 0.8 , indicating imperfect shell resolution.

The FFGFT T^4 graph shows no evaluable shells (regular grid, no shell separation at this threshold). This is a property of the regular T^4 grid, not an error.

Result Layer 3: Numerical Verification

$$k^* = \frac{\log \varphi}{\xi} \approx 3609.1, \quad r(3609) = 0.6179940013 \dots \approx \frac{1}{\varphi} \quad (|\text{error}| = 4.0 \times 10^{-5}, \text{ relative } 6.5 \times 10^{-5}). \quad (8)$$

The result follows without free parameters from ξ and φ ; the error statement was made precise in v3 (previously incorrectly $< 10^{-9}$).

.6 Discussion

Demarcation from the φ^N -triviality warning (Doc. 190 P35)

Doc. 272 (and Doc. 190 P35, ξ^N -triviality) state a methodological guardrail: *Every quantity can be written as φ^N ; a measured scale in an exponent is not a prediction*. This warning is justified and remains valid — but it does *not* apply to the k^* result of this document, for three reasons:

1. **No free exponent.** In a φ^N expression, N is a choice or a fit. Here $k^* = \log(\varphi)/\xi$ is a *result* — there is no freedom to choose k^* differently. The value 3609 follows necessarily from $\xi = 4/30000$.
2. **No empirical anchor.** Doc. 009 (P10 in Doc. 190) writes $245 = \varphi^{12}/(1.314)$ with an empirical factor 1.314. That is a genuine φ^N case: the denominator is not derived. In the k^* result there is no such denominator — $r(0) = D_f/3$ is geometrically defined, not empirically calibrated.
3. **No external calibration.** The calculation contains only $\xi = 4/30000$ (definition quantity) and φ (mathematical constant). No SI unit choice, no mass scale, no renormalisation step.

The three-level separation of φ -roles documented in Doc. 190 P37 — (1) microscopic/ ξ -level, (2) macroscopic/ $1/\varphi$ as fixed point, (3) particle physics/ Q_{FFGFT} — is implemented throughout this document. Doc. 188 already uses φ^{-1} as a phenomenological scaling factor between fractal levels; the present document shows *why* this factor appears structurally: it is the limit of the ξ -recursion, not an independent assumption.

Position in the Document Series

Doc. 270 describes the reduction chain $T^4 \rightarrow T^0$ as two qualitatively different operation types. The present document adds the *scale hierarchy*: the $q = 2/3$ recursion connects the microscopic ξ -level with the macroscopic φ -level through cumulative geometric damping — without a qualitative change in operation type.

Doc. 271 provides a structural comparison of FFGFT and HLV. The present document makes the spectral comparison announced there concrete through the numerical script and identifies `gap_CV` as a shared diagnostic channel.

Open Points

1. Shell-structure discrimination ($1/\varphi$ vs. Q_{FFGFT}) requires $N \gtrsim 2000$ points.
2. FFGFT- T^4 Laplacian on a finite graph (not only analytic) for a more direct comparison with HLV.
3. Pre-declared ensemble band tests analogous to Krüger Run E for FFGFT T^4 are still missing.
4. Physical observables that would make $k^* \approx 3609$ as a scale transition measurable — open.

.7 Conclusion

1. $1/\varphi$ is the fixed point of the ξ -damping recursion after $k^* = \log(\varphi)/\xi \approx 3609$ geometric scale levels with factor $q = 2/3$ — no free parameter.
2. With $q = 2/3$, $(2/3)^{3609}$ is convergent and internally consistent, not an astronomical limit.
3. **New result (v3, $N = 2000$):** FFGFT T^4 finite graph lies outside the null band with `gap_CV` = 1.68 ± 0.09 (band upper bound 1.05). HLV lies inside the band with `gap_CV` = $0.755 \approx$ jittered-cubic = 0.755 ($\Delta < 0.001$). HLV is spectrally indistinguishable from jittered-cubic controls at $N = 2000$ — consistent with Krüger Run E.
4. Shell structure at $N = 2000$: $Q_{\text{FFGFT}} \approx 0.6677$ fits the HLV shell structure better than $1/\varphi \approx 0.6180$ ($\Delta = 0.145$). Preliminary finding; absolute errors still large.
5. The three scalars ξ , $1/\varphi$, Q_{FFGFT} are all derivable but not interchangeable — they operate at different physical levels (Doc. 190 P37).

Connections. Doc. 258 (Q_{FFGFT} , Koide ladder), Doc. 270 (reduction chain $T^4 \rightarrow T^0$), Doc. 271 (FFGFT-HLV structural comparison), Doc. 272 (HLV/Dot-Theory context). The Python script `ffgft_hlv_gap_v2.py` is publicly available and reproducible.

Bibliography

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